

VOIDSCOPE · REQUIREMENTS

VoidScope requirements

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DOCUMENT CONTROL

Document ID	VDS-REQ-001	Version	0.3
Status	Draft	Date	2026-10-03
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/voidscope		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-03	Amish Chadha	TRL 3: status from VDS-CAL-001 on the constructable design (VDS-DDR-002)
0.2	2026-10-03	Amish Chadha	TRL 2: R2 split into reach (R2) and bends (R11); R5 restated as a ceiling with a bite valve; R6 adds the blocked-tip limit; R8 worded as a value-engineering target; R12 two-way voice added; concept status for each
0.1	2026-09-30	Amish Chadha	Initial scaffold

VoidScope requirements

Ten of the twelve requirements are met on paper by the constructable design: seven by calculation and three by design, all to be confirmed by test at TRL 4. R9 (setup in 3 minutes) is estimated at 2.1 minutes but can only be settled by timed drills. R11, following a path with two 45° bends, is not met: the rigid rod of the first version goes straight only.

Table 1. Requirements and status at TRL 3 (tags in brackets refer to VDS-CAL-001).

ID	Requirement	Target	Verification (TRL 4 or later)	Status at TRL 3
R1	Probe passes a standard core hole	Outer diameter of tip and rod at or below 50 mm	Pass the assembled probe through a 51 mm test ring along its full length	Met: 48 mm at the head, 47 mm at the grip [A1]
R2	Reach into a void along a straight path	Working length of at least 4 m (13 ft)	Measure assembled length from the grip to the tip	Met: 4,063 mm [A4]
R3	See in darkness and dust	Read a 20 mm high letter at 1 m in a dark, dusty test box	Test box with a set dust load and no ambient light	Met on paper: 19 pixels per letter against 10 needed; 7 lx left after 2 m of dusty air [C2, C6]; dust load to test
R4	Survive rubble handling	Head survives 20 drops of 1 m onto concrete and 1 m water immersion for 30 minutes	Drop and immersion tests followed by a video check	By design: IP68 camera, aluminium head with 2.85 mm minimum wall [G2]; to test
R5	Controlled water delivery	Drip from 0 up to a ceiling of no more than 5 mL/min at any allowed height up to 40 °C; no flow when the clamp is closed or the valve is not bitten	Timed collection at the tip at set heights and temperatures	Met: ceiling 1.5 mL/min at 1 m and 20 °C, 4.5 mL/min at 2 m and 40 °C [D1, D2]
R6	Air delivery to the void	At least 20 L/min of filtered air at the tip; never more than 1 kPa above ambient, even with the tip blocked	Flow meter and manometer at the tip, open and blocked	Met: 44 L/min at full speed, 0.9 kPa at shut-off [E3, E5]
R7	Shift-long battery life	At least 4 h of continuous video and light at 20 °C and at -10 °C	Run-down test at 20 °C and at -10 °C	Met: 6.2 h and 4.3 h on one pack [F4]
R8	Cost against the value-engineering target	Full kit parts cost against the USD 2,000 target	Costed bill of materials from named suppliers	USD 1,578, USD 422 under the target [I2, I4]
R9	Quick to deploy	Two trained people assemble and start searching within 3 minutes	Timed drills with a USAR partner	Estimate 2.1 min [H5]; only a timed drill can settle it
R10	Hygienic water path	Water path parts replaceable in under 5 minutes without tools	Timed replacement against a cleaning procedure	By design: single-use water set, about 4 min [D8]; to time
R11	Follow a bent path	Pass a 4 m test channel with two 45° bends	Push through the channel	Not met. The rigid rod cannot bend [A6]; a flexible lead section is a later version
R12	Talk with the survivor	Two-way voice between the operator and a person at the tip	Speech test in a rubble box	By design: tip speaker as speaker and microphone, push-to-talk headset

■ Assumptions

- Local crews can open or find gaps of about 50 mm into most voids of interest.

- A rigid, sectioned rod is good enough for a first version; articulation and bends come later (R11).
- A slow gravity drip through a bite valve is an acceptable way to give water when a medical lead directs it; the medical lead's acceptance is a TRL 4 item.
- Commodity sealed pipe-inspection camera heads meet the image, light and sealing targets.

Safety: These requirements describe rescue equipment that delivers water and air to a trapped person and carries a lithium battery. Meeting them on paper does not make the design safe to use. VoidScope is an open engineering reference, not certified rescue equipment and not a medical device; see VDS-PRC-001, Safety.